

Arctic tundra

The cold, almost treeless tundra is the zone of the Arctic lying between the northern forests and the polar ice. The landscape is frozen in winter because there is virtually no sunlight, but it thaws at the surface in the short summer, allowing plants to grow and insects to hatch. These attract animals from farther south, which leave again in winter.

FROZEN SWAMPS

Over most of the tundra, the ground is permanently frozen below the surface. This layer of “permafrost” is waterproof, so although the ground surface thaws in summer, the meltwater cannot drain away. This creates vast areas of waterlogged peat dotted with bog pools, which form some of the largest swamps in the world. These freeze solid again when the temperature drops in winter.

FACT!

The thick white winter coat of the Arctic fox is so efficient that the fox only starts shivering if the temperature falls below -94°F (-70°C).

▼ BLEAK LANDSCAPE

Trees are scarce on tundra because of the harsh climate and the way the ground keeps thawing and flooding.



SPRING THAW

When the Sun reappears in spring, it warms the ground, thawing the top layer of frozen earth. This gives the tundra plants their chance to grow, flower, and disperse their seeds in the patches of land that are not waterlogged. They are mostly very low-growing types that can resist strong, freezing winds often loaded with ice crystals. They include cotton grass (left), whose fluffy heads insulate the seeds against the cold, and miniature trees such as dwarf willow and dwarf birch.